

Power Budget

Team Number:	TBD
Project Name:	Rover Environmental Sensor PCB
Team Member Names:	TBD
Version:	v1.0 (2026-03-10)

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements						
All Major Components	Component Name	Part Number	Supply Voltage Range	#	Rated Maximum Current (mA)	Unit
	PIC18F57Q43 Microcontroller	PIC18F57Q43-I/PT	1.8V - 5.5V	1	200.00	200.00 mA
	MCP9808 Temperature Sensor	MCP9808T-E/MC	2.7V - 5.5V	1	0.40	0.40 mA
					0	0 mA
					0	0 mA
					0	0 mA
					0	0 mA
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.						
+12V Power Rail (unused)	Component Name	Part Number	Supply Voltage Range	#	Rated Maximum Current (mA)	Unit
					0	0 mA
					0	0 mA
					0	0 mA
					0	0 mA
					0	0 mA
					0	0 mA
					Subtotal	0 mA
					Safety Margin	25%
					Total Current Required on +12V Rail	0 mA
c1. Regulator or Source Chip	N/A	N/A	N/A	1	0	0 mA
					Total Remaining Current Available on +12V Rail	0 mA
+5V Power Rail (unused)	Component Name	Part Number	Supply Voltage Range	#	Rated Maximum Current (mA)	Unit
					0	0 mA
					0	0 mA
					0	0 mA
					0	0 mA
					0	0 mA
					0	0 mA
					Subtotal	0 mA
					Safety Margin	25%
					Total Current Required on +5V Rail	0 mA
c2. Regulator or Source Chip	N/A	N/A	N/A	1	0	0 mA
					Total Remaining Current Available on +5V Rail	0 mA
-5V Power Rail (unused)	Component Name	Part Number	Supply Voltage Range	#	Rated Maximum Current (mA)	Unit
					0	0 mA
					0	0 mA
					0	0 mA
					0	0 mA

						Subtotal	0	mA					
						Safety Margin	25%						
						Total Current Required on -5V Rail	0	mA					
c3. Regulator or Source Ch	N/A	N/A	N/A	1	0		0	mA					
						Total Remaining Current Available on -5V Rail						0	mA
+3.3V Power Rail													
Component Name		Part Number	upplyVoltageRange	#	uteMaximumCurrent (mA)	nalCurrent(mA)	Unit						
PIC18F57Q43 Microcontroller		PIC18F57Q43-I/PT	1.8V - 5.5V	1	200.00	200.00	mA						
MCP9808 Temperature Sensor		MCP9808T-E/MC	2.7V - 5.5V	1	0.40	0.40	mA						
						0	mA						
						0	mA						
						Subtotal	200.4	mA					
						Safety Margin	25%						
						Total Current Required on +3.3V Rail	250.5	mA					
c4. Regulator or Source Ch	+3.3V buck regulator	LM2595S-3.3/NOPB	4.5V - 40V	1	1000.00		1000.00	mA					
						Total Remaining Current Available on 3.3V Rail						749.5	mA
C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining													
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails													
External Power Source 1 (s		Component Name	Part Number	upplyVoltageRange	Output	uteMaximumCurrent (mA)	nalCurrent(mA)	Unit					
Power Source 1 Selection		12V Wall Supply	SWI24-12-N-P5	90VAC - 264VAC	+12V	2000.00	2000.00	mA					
Power Rails Connected to External Power Source 1		+3.3V buck regulator	LM2595S-3.3/NOPB	4.5V - 40V	1	1000	1000	mA					
							0	mA					
							0	mA					
							1000	mA					
Total Remaining Current Available on External Power Source 1													
External Power Source 2 (u		Component Name	Part Number	upplyVoltageRange	Output	uteMaximumCurrent (mA)	nalCurrent(mA)	Unit					
Power Source 2 Selection							100	mA					
Power Rails Connected to External Power Source 2							0	mA					
							0	mA					
							0	mA					
							100	mA					
Total Remaining Current Available on External Power Source 2													
E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in													
		Component Name	Part Number	upplyVoltageRange	Capacity(mAh)	iredByRegulators							

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit